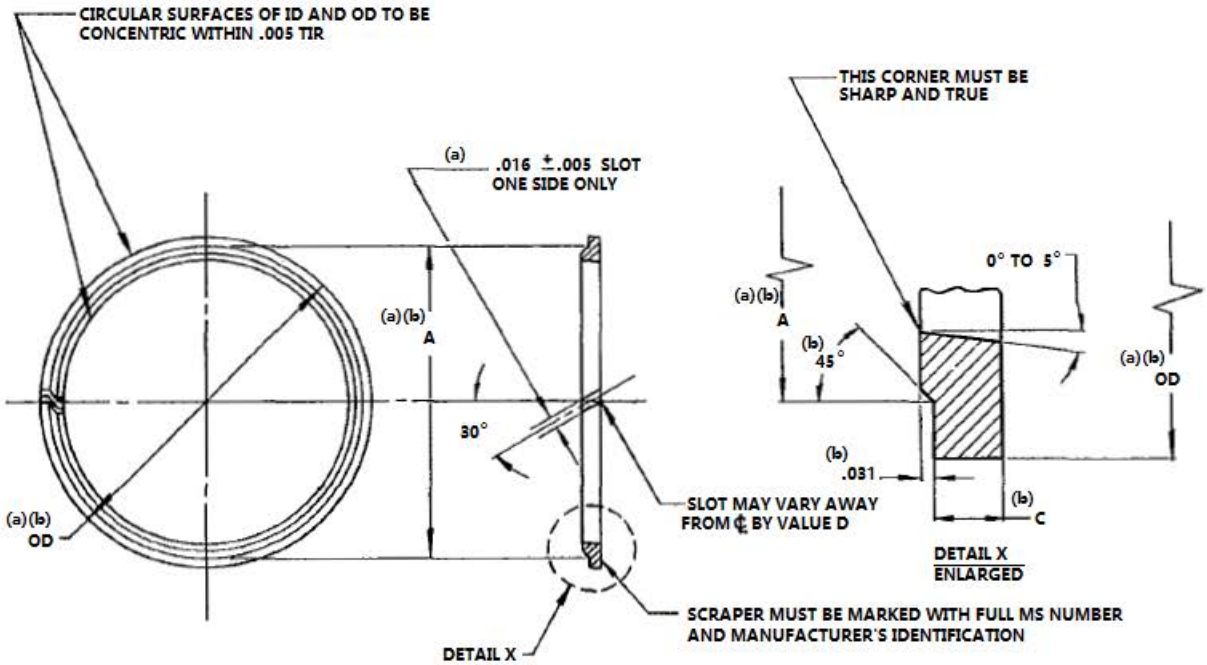
FED. SUP CLASS
1650



(a) DIMENSIONS TO BE TAKEN WITH SCRAPER INSTALLED ON HIGH TOLERANCE ROD AS SPECIFIED IN PROCUREMENT SPECIFICATION.

(b) APPLICABLE TO ALL METAL TYPE ONLY. RUBBER LOADED TYPE MAY DEVIATE FROM THESE DIMENSIONS AS SPECIFIED ON MANUFACTURER'S DRAWINGS APPROVED AS PART OF QUALIFICATION DATA.

MATERIAL: SEE PROCUREMENT SPECIFICATION.

FINISH: CADMIUM PLATE, QQ-P-416, CLASS 2, TYPE II, ALL SCRAPERS FABRICATED OF COPPER BASE ALLOYS. SURFACE FACING THE PISTON ROD MAY BE UNPLATED. ALL OTHER SURFACES SHALL BE PLATED AND THE FINISHED SCRAPER SHALL OTHERWISE BE IN ACCORDANCE WITH THE PROCUREMENT SPECIFICATION.

FOR A DISTANCE OF .031 INCHES ON EACH SIDE OF THE SCRAPING EDGE, TOOL MARKS UP TO 50√ MAXIMUM WILL BE ACCEPTABLE. THE REST OF THE SCRAPER MAY HAVE TOOL MARKS UP TO 100√.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.016, ANGLES ±1/2°

INSTALLATION IN ACCORDANCE WITH MS33675.

EXAMPLES OF PART NUMBERS: MS28776M2-21 = SCRAPER FOR 2 INCH ROD, ALL METAL, -65° TO +275°F.
MS28776R2-21 = SCRAPER FOR 2 INCH ROD, WITH RUBBER, -65° TO +275°F.

(D) NOTE: THE NUMBER 2 AFTER M OR R DESIGNATES A CLASS 2 TEMPERATURE RANGE. (SEE MIL-R-5049).

ALL UNITS REFERENCED IN THIS DOCUMENT CAN BE EITHER SINGLE OR MULTI-PIECE CONSTRUCTION.

NOTE: IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN; THE TEST OF THIS STANDARD SHALL TAKE PRECEDENCE.

FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

APPROVED 26 Apr 57 REVISED (A) 3 Jul 63 (B) 6 Apr 73 (C) 7 Sep 87 (D) 15 Jun 1988

REVIEWER ACTIVITIES: ARMY - MI
USER ACTIVITIES: ARMY -ME

DISTRIBUTION STATEMENT A.
AMSC N/A

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P.A. Air Force - 71 Other Cust Army - AV Navy - AS AF - 99	TITLE **SCRAPER, PISTON ROD**	**MILITARY STANDARD** **MS28776**
PROCUREMENT SPECIFICATION (D) MIL-R-5049	SUPERSEDES: MS 28903	SHEET 1 OF 2

DD FORM 1 Sep 97 672-1 (Coordinated)

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FED. SUP CLASS 1650

DASH NO.	NOMINAL ROD OD	(a)(b) SCRAPER OD +.000 -.005	(a)(b) A +/- .005 DIA	(b) C +.000 -.010	D MAX	BREAKOUT FRICTION LB MAX
1	1/2	.740	.617	.094	.050	4
2	9/16	.803	.680			
3	5/8	.865	.742			
4	11/16	.928	.804			
5	3/4	.990	.867			
6	13/16	1.032	.919			
7	7/8	1.094	.982			
8	15/16	1.157	1.044			
9	1	1.219	1.106			
10	1- 1/6	1.282	1.169			9
11	1- 1/8	1.344	1.232			
12	1- 3/16	1.407	1.294			
13	1- 1/4	1.469	1.356			
14	1- 5/16	1.594	1.450			
15	1- 3/8	1.657	1.512			
16	1- 7/16	1.719	1.575			
17	1- 1/2	1.782	1.638			
18	1- 5/8	1.907	1.763			
19	1- 3/4	2.032	1.888			
20	1- 7/8	2.157	2.013			
21	2	2.282	2.138		.093	
22	2- 1/8	2.407	2.263			
23	2- 1/4	2.532	2.388			
24	2- 3/8	2.657	2.513			
25	2- 1/2	2.782	2.638			
26	2- 5/8	2.969	2.794	.109		12
27	2- 3/4	3.094	2.919			
28	2- 7/8	3.219	3.044			
29	3	3.344	3.169			14
30	3- 1/8	3.469	3.294			
31	3- 1/4	3.594	3.419			
32	3- 3/8	3.719	3.544			
33	3- 1/2	3.844	3.669			
34	3- 5/8	3.969	3.794			
35	3- 3/4	4.094	3.919			
36	3- 7/8	4.219	4.044		.125	22
37	4	4.407	4.200	.125		
38	4- 1/8	4.532	4.325			
39	4- 1/4	4.657	4.450			
40	4- 3/8	4.782	4.575			
41	4- 1/2	4.907	4.700			
42	4- 5/8	5.032	4.825			30
43	4- 3/4	5.157	4.950			
44	4- 7/8	5.282	5.075			
45	5	5.407	5.200			
46	5- 1/8	5.532	5.325			40
47	5- 1/4	5.657	5.450			
48	5- 3/8	5.782	5.575			
49	5- 1/2	5.907	5.700			
50	5- 5/8	6.094	5.856	.141	.250	50
51	5- 3/4	6.219	5.982			
52	5- 7/8	6.344	6.106			
53	6	6.469	6.232			
54	6- 1/4	6.719	6.482			
55	6- 1/2	6.969	6.732			
56	6- 3/4	7.219	6.982			
57	7	7.469	7.232			
58	7- 1/4	7.719	7.482			
59	7- 1/2	7.969	7.732			
60	7- 3/4	8.219	7.982			
61	8	8.469	8.232			
62	8- 1/2	8.969	8.732			70
63	9	9.469	9.232			
64	9- 1/2	9.969	9.732			
65	10	10.469	10.232			
66	10- 1/2	10.969	10.732			
67	11	11.469	11.232			
68	11- 1/2	11.969	11.732	.156		
69	12	12.469	12.232			
70	12- 1/2	12.969	12.732			
71	13	13.469	13.232			

APPROVED 26 Apr 1957 REVISED Ⓓ FOR CHANGES SEE SHEET 1. 15 JUNE 1988

P.A. Air Force - 71
Other Cust. Army - AV
Navy - AS
AF - 99

PROCUREMENT SPECIFICATION
Ⓓ MIL-R-5049

TITLE

SCRAPER, PISTON ROD

MILITARY STANDARD

MS28776

SUPERSEDES: MS28903

SHEET 2 OF 2

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